In [9]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [15]:
```
def f(x):
    return x**2 + 6*x + 9
```

In [16]:
```
x = 2
lr = 0.01

for i in range(100):
    x = x - lr * df(x)
    print("Iteration", i+1, ": x =", round(x, 4),
          ", f(x) =", round(df(x), 4))
```

```
Iteration 1 : x = 1.9 , f(x) = 9.8
Iteration 2 : x = 1.802 , f(x) = 9.604
Iteration 3 : x = 1.706 , f(x) = 9.4119
Iteration 4 : x = 1.6118 , f(x) = 9.2237
Iteration 5 : x = 1.5196 , f(x) = 9.0392
Iteration 6 : x = 1.4292 , f(x) = 8.8584
Iteration 7 : x = 1.3406 , f(x) = 8.6813
Iteration 8 : x = 1.2538 , f(x) = 8.5076
Iteration 9 : x = 1.1687 , f(x) = 8.3375
Iteration 10 : x = 1.0854 , f(x) = 8.1707
Iteration 11 : x = 1.0037 , f(x) = 8.0073
Iteration 12 : x = 0.9236 , f(x) = 7.8472
Iteration 13 : x = 0.8451 , f(x) = 7.6902
Iteration 14 : x = 0.7682 , f(x) = 7.5364
Iteration 15 : x = 0.6928 , f(x) = 7.3857
Iteration 16 : x = 0.619 , f(x) = 7.238
Iteration 17 : x = 0.5466 , f(x) = 7.0932
Iteration 18 : x = 0.4757 , f(x) = 6.9514
Iteration 19 : x = 0.4062 , f(x) = 6.8123
Iteration 20 : x = 0.338 , f(x) = 6.6761
Iteration 21 : x = 0.2713 , f(x) = 6.5426
Iteration 22 : x = 0.2059 , f(x) = 6.4117
Iteration 23 : x = 0.1417 , f(x) = 6.2835
Iteration 24 : x = 0.0789 , f(x) = 6.1578
Iteration 25 : x = 0.0173 , f(x) = 6.0346
Iteration 26 : x = -0.043 , f(x) = 5.914
Iteration 27 : x = -0.1022 , f(x) = 5.7957
Iteration 28 : x = -0.1601 , f(x) = 5.6798
Iteration 29 : x = -0.2169 , f(x) = 5.5662
Iteration 30 : x = -0.2726 , f(x) = 5.4548
Iteration 31 : x = -0.3271 , f(x) = 5.3457
Iteration 32 : x = -0.3806 , f(x) = 5.2388
Iteration 33 : x = -0.433 , f(x) = 5.1341
Iteration 34 : x = -0.4843 , f(x) = 5.0314
Iteration 35 : x = -0.5346 , f(x) = 4.9307
Iteration 36 : x = -0.5839 , f(x) = 4.8321
Iteration 37 : x = -0.6323 , f(x) = 4.7355
Iteration 38 : x = -0.6796 , f(x) = 4.6408
Iteration 39 : x = -0.726 , f(x) = 4.548
Iteration 40 : x = -0.7715 , f(x) = 4.457
Iteration 41 : x = -0.8161 , f(x) = 4.3679
Iteration 42 : x = -0.8597 , f(x) = 4.2805
Iteration 43 : x = -0.9026 , f(x) = 4.1949
Iteration 44 : x = -0.9445 , f(x) = 4.111
Iteration 45 : x = -0.9856 , f(x) = 4.0288
Iteration 46 : x = -1.0259 , f(x) = 3.9482
Iteration 47 : x = -1.0654 , f(x) = 3.8692
Iteration 48 : x = -1.1041 , f(x) = 3.7919
Iteration 49 : x = -1.142 , f(x) = 3.716
Iteration 50 : x = -1.1792 , f(x) = 3.6417
Iteration 51 : x = -1.2156 , f(x) = 3.5689
Iteration 52 : x = -1.2513 , f(x) = 3.4975
Iteration 53 : x = -1.2862 , f(x) = 3.4275
Iteration 54 : x = -1.3205 , f(x) = 3.359
Iteration 55 : x = -1.3541 , f(x) = 3.2918
Iteration 56 : x = -1.387 , f(x) = 3.226
Iteration 57 : x = -1.4193 , f(x) = 3.1614
Iteration 58 : x = -1.4509 , f(x) = 3.0982
Iteration 59 : x = -1.4819 , f(x) = 3.0363
Iteration 60 : x = -1.5122 , f(x) = 2.9755
```

```
Iteration 61 : x = -1.542 , f(x) = 2.916
Iteration 62 : x = -1.5711 , f(x) = 2.8577
Iteration 63 : x = -1.5997 , f(x) = 2.8005
Iteration 64 : x = -1.6277 , f(x) = 2.7445
Iteration 65 : x = -1.6552 , f(x) = 2.6896
Iteration 66 : x = -1.6821 , f(x) = 2.6359
Iteration 67 : x = -1.7084 , f(x) = 2.5831
Iteration 68 : x = -1.7343 , f(x) = 2.5315
Iteration 69 : x = -1.7596 , f(x) = 2.4808
Iteration 70 : x = -1.7844 , f(x) = 2.4312
Iteration 71 : x = -1.8087 , f(x) = 2.3826
Iteration 72 : x = -1.8325 , f(x) = 2.3349
Iteration 73 : x = -1.8559 , f(x) = 2.2883
Iteration 74 : x = -1.8788 , f(x) = 2.2425
Iteration 75 : x = -1.9012 , f(x) = 2.1976
Iteration 76 : x = -1.9232 , f(x) = 2.1537
Iteration 77 : x = -1.9447 , f(x) = 2.1106
Iteration 78 : x = -1.9658 , f(x) = 2.0684
Iteration 79 : x = -1.9865 , f(x) = 2.027
Iteration 80 : x = -2.0068 , f(x) = 1.9865
Iteration 81 : x = -2.0266 , f(x) = 1.9468
Iteration 82 : x = -2.0461 , f(x) = 1.9078
Iteration 83 : x = -2.0652 , f(x) = 1.8697
Iteration 84 : x = -2.0839 , f(x) = 1.8323
Iteration 85 : x = -2.1022 , f(x) = 1.7956
Iteration 86 : x = -2.1201 , f(x) = 1.7597
Iteration 87 : x = -2.1377 , f(x) = 1.7245
Iteration 88 : x = -2.155 , f(x) = 1.69
Iteration 89 : x = -2.1719 , f(x) = 1.6562
Iteration 90 : x = -2.1884 , f(x) = 1.6231
Iteration 91 : x = -2.2047 , f(x) = 1.5906
Iteration 92 : x = -2.2206 , f(x) = 1.5588
Iteration 93 : x = -2.2362 , f(x) = 1.5277
Iteration 94 : x = -2.2514 , f(x) = 1.4971
Iteration 95 : x = -2.2664 , f(x) = 1.4672
Iteration 96 : x = -2.2811 , f(x) = 1.4378
Iteration 97 : x = -2.2955 , f(x) = 1.4091
Iteration 98 : x = -2.3096 , f(x) = 1.3809
Iteration 99 : x = -2.3234 , f(x) = 1.3533
Iteration 100 : x = -2.3369 , f(x) = 1.3262
```

In [17]:
```
minimum = x
print("Local minimum occurs at x =", round(minimum, 4),
      ", f(x) =", round(df(minimum), 4))
```

```
Local minimum occurs at x = -2.3369 , f(x) = 1.3262
```

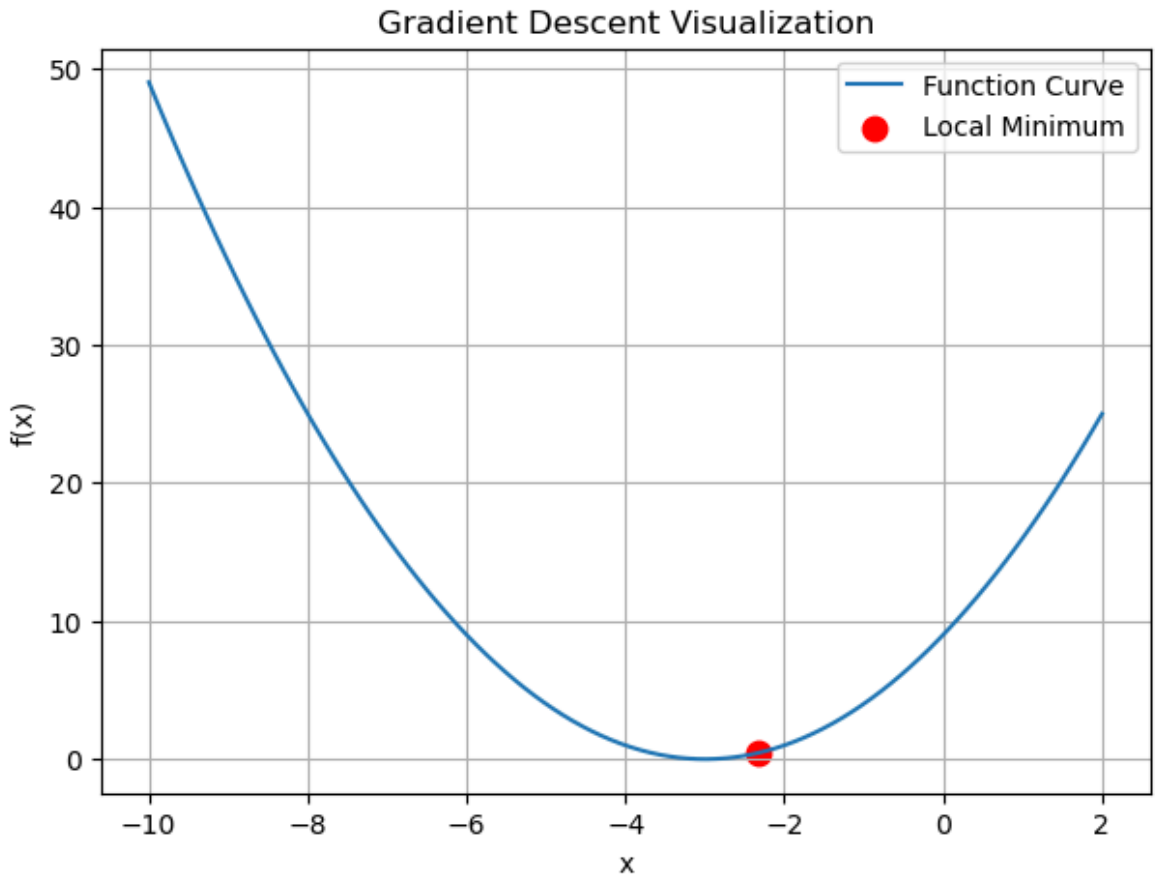
In [20]:
```
minimum = x

plt.figure(figsize=(7,5))
plt.plot(x_vals, y_vals, label="Function Curve")

plt.scatter(minimum, f(minimum), color='red',
            s=80, label="Local Minimum")

plt.xlabel("x")
plt.ylabel("f(x)")
plt.title("Gradient Descent Visualization")
plt.legend()
```

```
plt.grid()
plt.show()
```



In [ ]: